

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5115

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I Nissanki Sri Venkata Lakshmi Kavya (192373036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N.S.V.L.Kavya

Annexure A

Analytical Problem Solving

4. Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23$, $g=5$
- b. User A private key = 6
- c. User B private key = 15
- d. Compute public keys for A and B
- e. Compute the shared secret key
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- g. Suggest a secure method to prevent the attack

Given:

- Prime number: $p = 23$
- Generator: $g = 5$
- User A private key: $a = 6$
- User B private key: $b = 15$

(d) Compute the Public Keys of A and B

Formula:

- Public Key of A:

$$A = g^a p$$

- Public Key of B:

$$B = g^b p$$

Public Key of A

$$\begin{aligned} A &= 5^6 23 \\ 5^6 &= 15625 \\ 15625 23 &= 8 \end{aligned}$$

Public Key of A = 8

Public Key of B

$$B = 5^{15} 23$$

Using modular exponentiation:

$$5^{15} 23 = 19$$

Public Key of B = 19

(e) Compute the Shared Secret Key

At User A:

$$\begin{aligned} K &= B^a p \\ K &= 19^6 23 = 2 \end{aligned}$$

At User B:

$$\begin{aligned} K &= A^b p \\ K &= 8^{15} 23 = 2 \end{aligned}$$

Shared Secret Key = 2

(f) Man-in-the-Middle (MITM) Attack

Suppose an attacker (**M**) intercepts the communication.

1. A sends public key **8** to B.
2. The attacker intercepts it and sends a fake public key (say **10**) to B.
3. B sends public key **19** to A.
4. The attacker intercepts it and sends another fake public key (say **7**) to A.
5. A computes a secret key with the attacker instead of B.
6. B also computes a different secret key with the attacker.
7. The attacker now decrypts, reads, modifies, and re-encrypts all messages between A

and B without their knowledge.

Result: A and B believe they are communicating securely, but the attacker can read and alter all communications.

(g) Secure Method to Prevent the Attack

To prevent a Man-in-the-Middle attack, use **authentication** along with Diffie–Hellman.

Recommended methods:

- **Digital Certificates (PKI)**
- **Digital Signatures**
- **TLS/SSL Protocol**
- **RSA or ECC-based authentication**

Conclusion:

Using **Digital Certificates and TLS/SSL** verifies the identity of both users, ensuring that public keys are authentic and preventing Man-in-the-Middle attacks.

5. ECC vs RSA Comparison

Given:

- a. RSA key size = 1024 bits
- b. ECC key size = 160 bits (equivalent security)
- c. Device processing capacity = limited (IoT device)
- d. Analyze computational cost for both systems
- e. Estimate which consumes less power and memory
- f. Justify which cryptosystem is better for the given scenario with reasoning

Given:

- RSA key size = 1024 bits
- ECC key size = 160 bits (equivalent security)
- Device = IoT device (limited processing capacity)

(d) Analyze Computational Cost for Both Systems

RSA	ECC
Uses large key size (1024 bits).	Uses small key size (160 bits).
Requires more CPU processing for encryption/decryption.	Requires fewer computations, making it faster.
Higher computational cost.	Lower computational cost.

Answer:

- **RSA:** High computational cost due to larger key size.
- **ECC:** Low computational cost due to smaller key size.

(e) Estimate Which Consumes Less Power and Memory

RSA	ECC
Higher memory usage.	Lower memory usage.
Consumes more battery power.	Consumes less battery power.
Needs more storage for	Requires less storage.

keys.	
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Answer:

ECC consumes less power and memory because it uses much smaller keys while providing the same level of security as RSA.

(f) Justify Which Cryptosystem is Better for the Given Scenario

For an **IoT device**, processing power, memory, and battery life are limited.

ECC is the better choice because:

- Provides the same security as RSA with much smaller keys (160-bit ECC $\approx$ 1024-bit RSA).
- Faster encryption and decryption.
- Lower computational cost.
- Uses less memory and storage.
- Consumes less power, increasing battery life.
- Well suited for resource-constrained devices such as IoT sensors, smart watches, and embedded systems.

Conclusion:

ECC (Elliptic Curve Cryptography) is the best cryptosystem for the given IoT scenario because it offers equivalent security with lower computational cost, less memory usage, and lower power consumption than RSA.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

strong
justification

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation